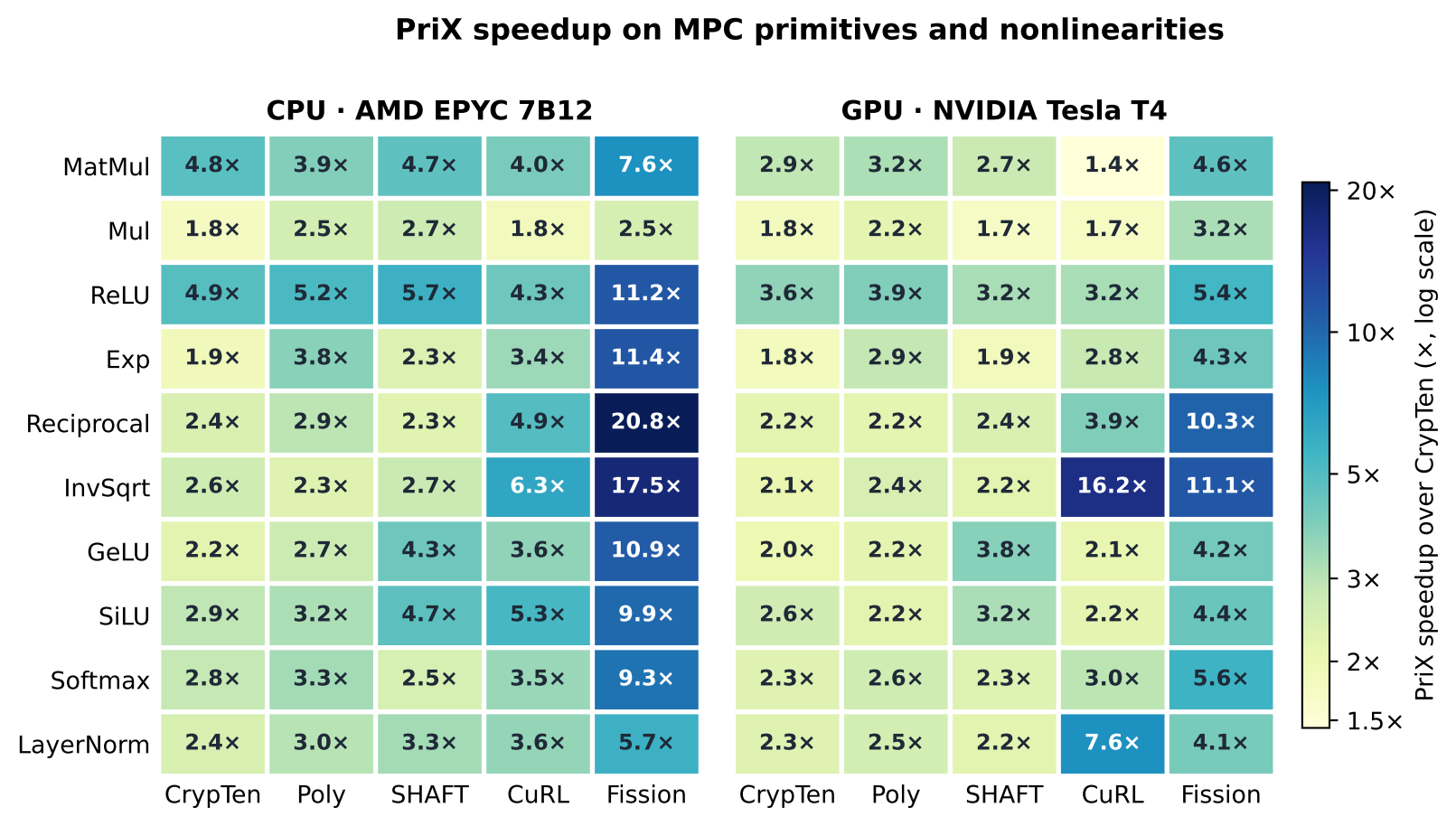
PriX speedup on MPC primitives and nonlinearities
CPU · AMD EPYC 7B12
GPU · NVIDIA Tesla T4
MatMul 4.8× 3.9× 4.7× 4.0× 7.6× 2.9× 3.2× 2.7× 1.4× 4.6×
Mul 1.8× 2.5× 2.7× 1.8× 2.5× 1.8× 2.2× 1.7× 1.7× 3.2×
ReLU 4.9× 5.2× 5.7× 4.3× 11.2× 3.6× 3.9× 3.2× 3.2× 5.4×
Exp 1.9× 3.8× 2.3× 3.4× 11.4× 1.8× 2.9× 1.9× 2.8× 4.3×
Reciprocal 2.4× 2.9× 2.3× 4.9× 20.8× 2.2× 2.2× 2.4× 3.9× 10.3×
InvSqrt 2.6× 2.3× 2.7× 6.3× 17.5× 2.1× 2.4× 2.2× 16.2× 11.1×
GeLU 2.2× 2.7× 4.3× 3.6× 10.9× 2.0× 2.2× 3.8× 2.1× 4.2×
SiLU 2.9× 3.2× 4.7× 5.3× 9.9× 2.6× 2.2× 3.2× 2.2× 4.4×
Softmax 2.8× 3.3× 2.5× 3.5× 9.3× 2.3× 2.6× 2.3× 3.0× 5.6×
LayerNorm 2.4× 3.0× 3.3× 3.6× 5.7× 2.3× 2.5× 2.2× 7.6× 4.1×
CrypTen Poly SHAFT CuRL Fission CrypTen Poly SHAFT CuRL Fission
20×
10×
5×
3×
2×
1.5×
PriX speedup over CrypTen (×, log scale)